README — Multi-layer / Multi-input `core_tb`

Overview

This directory contains an updated `core_tb.v` that supports **multiple layers** and **multiple input images** (multi-case verification).
To use it, you will:

1. Replace the original `core_tb.v` in Part2.
2. Add a `config.txt` under `datafiles/` to describe the test set.
3. Generate matching `datafiles/` using `software/test.py`.

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1) Replace `core_tb.v`

Copy the `core_tb.v` from **this directory** and replace:

```
Part2_SIMD/hardware/verilog/core_tb.v
```

Then re-run your usual simulation command flow.

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2) Add `datafiles/config.txt`

Create this file:

```
Part2_SIMD/hardware/datafiles/config.txt
```

Example content:

```
2 4
36 16 6 4
36 16 6 4
```

Format explanation

- **Line 1**:

  ```
  <NUM_LAYER> <NUM_IMG>
  ```

  Example `2 4` means:

  - `NUM_LAYER = 2` layers (layer index: 0..1)
  - `NUM_IMG = 4` images per layer (img index: 0..3)

- **Line 2..(NUM_LAYER+1)**: one line per layer

```
<LEN_NIJ> <LEN_ONIJ> <A_PAD_NI_DIM> <O_NI_DIM>
```

Example `36 16 6 4` means:

- `LEN_NIJ = 36` (activation stream length; corresponds to 6×6 padded input positions)
- `LEN_ONIJ = 16` (output stream length; corresponds to 4×4 output positions)
- `A_PAD_NI_DIM = 6` (padded input width/height)
- `O_NI_DIM = 4` (output width/height)

Important constraint

Even though this `core_tb` can *parse* different per-layer sizes, **the hardware design constraints require the input/output format to remain**:

```
36 16 6 4
```

So all layers should use the same `36 16 6 4` in `config.txt`.

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3) Data file naming rules

All files live under:

```
Part2_SIMD/hardware/datafiles/<mode>/
```

Where `<mode>` is:

- `2bit` for SIMD / 2-bit mode
- `4bit` for normal / 4-bit mode

Activation files

```
activation_layer<layer>_img<img>_tile0.txt
```

Weight files (per layer and per kij)

```
weight_layer<layer>_itile0_otile0_kij<kij>.txt
```

Where `kij` ranges from `0` to `8` (3×3 kernel flattened).

Golden output (psum) files

```
psum_layer<layer>_img<img>.txt
```

Special case (compatibility with the original core_tb)

When **layer = 0** and **only one tile is used** (i.e., effectively a single-case setup), the naming is compatible with the original `core_tb`:

- activation / weight / psum filenames follow the same "single-layer/single-input" naming used by the original testbench.

(Use this when you want to drop-in replace without changing existing single-case datasets.)

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4) Generate test data

A generator script is provided:

```
software/test.py
```

How to run

1. Copy `software/test.py` into:

   ```
   Part2_SIMD/hardware/sims/test.py
   ```

2. Run it from `hardware/sims/`:

   ```
   python3 test.py
   ```

It will read:

```
../datafiles/config.txt
```

and generate all required activation / weight / psum files under:

```
../datafiles/2bit/
../datafiles/4bit/
```